

Your grade: 100%

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1. What step or steps does the IEKF iterate? (Select all that apply.)

1 / 1 point

- ☐ It execute the prediction steps multiple times to improve the state prediction.
- ☐ It executes the output measurement step (of z_k) multiple times to make a better averaged measurement of z_k .
- ☒ It iterates the measurement update steps multiple times to improve the state estimate.

✔

Correct

Yes. That is correct.

- ☐ It iterates the input measurement step (of u_k) multiple times to make a better averaged value of u_k .

2. Which equation is repeatedly re-linearized in the IEKF?

1 / 1 point

- ☐ The model state equation $f(\cdot)$.
- ☒ The model output equation $h(\cdot)$.
- ☐ The EKF equation that computes the Kalman gain L_k .
- ☐ The EKF equation that computes the state estimate based on the state prediction, the Kalman gain, and the innovation.
- ☐ The EKF equation that computes the estimation-error covariance matrix.

✔

Correct

Yes. That is correct.

3. In IEKF steps 1c-2c, which variables are recomputed for every iteration of the IEKF? (Select all that apply.)

1 / 1 point

- ☐ $\hat{x}_k^-$.
- ☒ $\hat{x}_k^+$.

✔

Correct

Yes. That is correct.

- ☐ $\Sigma_{\hat{x},k}^-$.
- ☒ $\Sigma_{\hat{x},k}^+$.

✔

Correct

Yes. That is correct.

- ☒ $\hat{C}_k$.

✔

Correct

Yes. That is correct.

- ☒ $\hat{D}_k$.

✔

Correct

Yes. That is correct.

- ☒ $\hat{z}_k$.

✔

Correct

Yes. That is correct.

- ☒ L_k .

✔

Correct

Yes. That is correct.